

RFID-Based Automatic Vehicle Parking System

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Radio-frequency identification (RFID) is an automatic identification method wherein the data stored on RFID tags or transponders is remotely retrieved. The RFID tag is a device that can be attached to or incorporated into a product, animal

or person for identification and tracking using radio waves. Some tags can be read from several metres away, beyond the line of sight of the reader.

RFID technology is used in vehicle parking systems of malls and buildings (refer Fig. 1). The system normally consists of a vehicle counter, sensors, display board, gate controller, RFID tags and RFID reader. Presented here is an automatic vehicle parking system using AT89S52 microcontroller.

RFID system fundamentals

Basically, an RFID system consists of an antenna or coil, a transceiver (with decoder) and a transponder (RF tag) electronically programmed with unique information. There are many different types of RFID systems in the market. These are categorised on the basis of their frequency ranges. Some of the most commonly used RFID kits are low-frequency (30-500kHz), mid-frequency (900kHz-1500MHz) and high-frequency (2.4-2.5GHz).

RFID antenna. Fig. 2 shows the internal diagram of a typical RFID antenna. The antenna emits radio signals to activate the tag and read/write data from/to it. It is the conduit between the tag and the transceiver, which controls the system's data acquisition

and communication.

Antennae are available in a variety of shapes and sizes. These can be built into a door frame to receive tag data from persons or things passing through the door, or mounted on an inter-state tollbooth to monitor the traffic passing by on a freeway. The electromagnetic field produced by the antenna can be constantly present when multiple tags are expected continually. If constant interrogation is not required, a sensor device can activate the field.

Often the antenna is packaged with a transceiver and decoder to act as a reader (interrogator), which can be configured either as a handheld or a fixed-mount device. The reader emits radio waves in the range of 2.5 cm to 30 metres or more, depending upon its power output and the radio frequency used. When an RFID tag passes through the electromagnetic zone, it detects the reader's activation signal. The reader decodes the data encoded in the tag's integrated circuit (silicon chip) and communicates to the host computer for processing.



Fig. 1: Automatic vehicle parking system

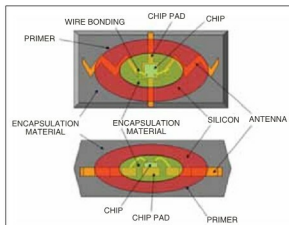


Fig. 2: Internal diagram of a typical RFID antenna

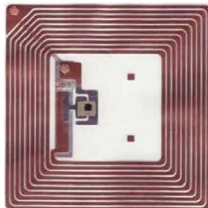


Fig. 3: Internal structure of typical RFID tag